DESIGN GROUP P40

Compressed Air Powered Vehicle

Department of Mechanical Engineering

Autumn Term – A.Y. 2023-2024

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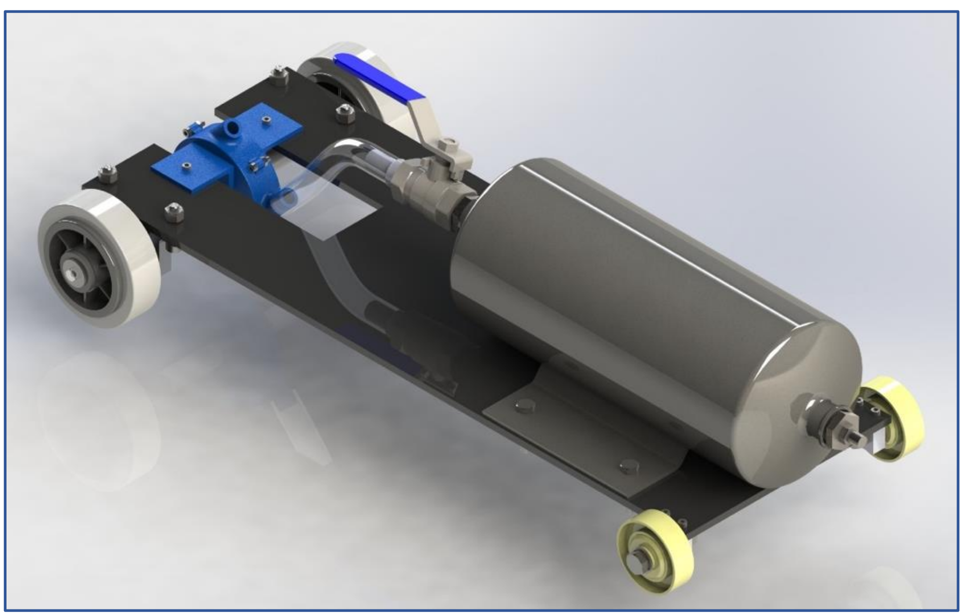


EXECUTIVE SUMMARY

The project aimed to design, build, and test a compressed-air vehicle with 100 hours of operation prior to failure. The power supply is a 2-litre compressed air reservoir at 6.5 bar. During the past 10 weeks, the group designed and manufactured a vehicle that will be raced in the City and Guilds building's concourse at Imperial College London. The purpose of this report is to outline key characteristics of the design process, discuss the technical decisions that were made during the development of the product, describe the manufacturing procedure, and present the results of the project. Starting with an overview of the design brief, identifying the main requirements and constraints and further developing them into a Product Design Specification (PDS).

The brief for this task included various constraints that the group must adhere to, for example the source of external components, officially chosen to be *RS Components*. Alongside these constraints, the group devised some key characteristics of the car that the final design would try to adhere to. These were a lightweight and simple design. Various design concepts were initially produced and analysed to reach the final and optimal solution. The final design consisted of a converging nozzle to control the airflow and spin the modified Pelton turbine. The transmission was chosen to be direct drive, mounting the turbine directly on the rear shaft.

The Functional and Operational description outlines the main steps to evaluate the relevant values and demonstrate the satisfaction of performance requirements. This includes calculations made, engineering drawings, sketches, and descriptions of key subsystems. The final design's critical features were identified and analysed in the Failure Modes Section through the calculations of Safety Factors to show that the design is safe under operational conditions.

The report also outlines the main manufacturing and assembling processes, showing a very efficient and high-quality approach to the realization of the product. Moving into a conclusion regarding the accomplishments of the project and considerations of improvement that could be made in the future. More details about the technical aspects of the project are shown in the Appendices where main Engineering Drawings, measurements report, and Project management are included.

The vehicle will be stored until the testing session that will be held during the 2024 Summer Term.

1. INTRODUCTION

The objective of this project was to design and manufacture a small compressed-gas car that can complete a 20 m straight track in the shortest time possible using 2L of gas at 6.5 bar. The pressure vessel (air canister), along with 2 front and 2 rear wheels were provided. The product was required to operate for a minimum of 100 hours before failure. RS Components was used as the supplier for power transmission parts, and the Student Training Workshop was used for manufacture.

The final design consists of a rear-drive vehicle powered by a turbine mounted directly on the shaft. The chassis develops in a length of 450 mm and a width of 250 mm. The vehicle is comprised of parts manufactured with a range of different manufacturing methods, including 3D printing, laser cutting, CNC, and milling. The design was successfully manufactured and assembled. The car is expected to travel the 20 m concourse in less than 3 seconds.

Each member was assigned specific responsibilities throughout the design and manufacturing process, as shown in Table 1.

Table 1: Team responsibilities

Team Member	Position
Matthew Morrissey	CAD Leader and Cloud Supervisor
Michael Pan	Design and Integration Leader
Gian Andrea Rossi	Project Manager
Grady Su	Project Secretary, and Testing and Verification Leader
Tanadon Tharahirunchot	Manufacturing Leader

2. DESIGN GATES

2.1 PDS

The version record and the final version of key design specifications are illustrated in the Product Design Specifications table (Table 2), based on the provided design brief (Mohammed, 2023).

Table 2: Product Design Specification

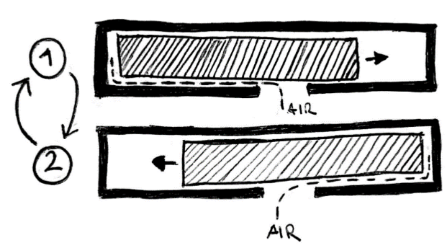
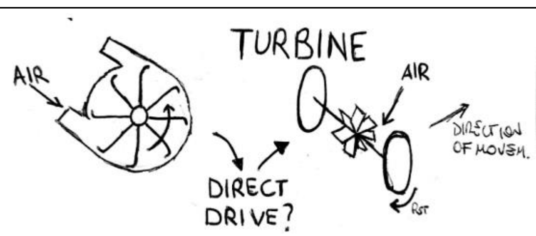
Version	Changes	Date
1.0	First draft	10/10/2023
1.1	Dimension and weight goals	14/10/2023
1.2	Product cost updated	1/11/2023
1.3	Added missing details on aesthetics	16/11/2023
Element	**Statement or criteria**	**Verification by**
Customer		
Needs	To develop a car powered by compressed air	Testing Session
Aesthetics	No exposed moving transmission elements	Design Tutor
Performance	The valve must be accessible Run in the CAGB concourse	Testing Session
Size	Width: target: 200-250 mm – max: 400mm Length: target: 450-500 mm – max: n.a. Height: target: 300 mm – max: 500mm	Design Tutor Design Gateway Tutor

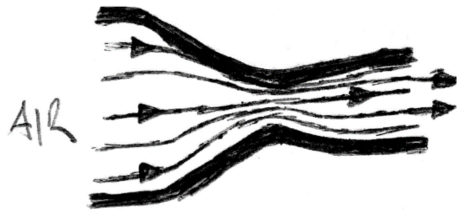
Weight	< 5 kg	STW Technicians
Life		
Service Life	100 hrs	Design Calculations
Producer		
Quantity	Single	Testing Session
Product cost	£205 (stock material and manufacturing excluded)	RS order form
Manufacturing Constraints	Max of three 3D printed parts with max dimensions 250 mm x 250 mm x 100 mm Transmission elements purchased from RS 1 CNC part, can be duplicated At least one part turned At least one part milled	STW Technicians Design Gateway Tutor RS order form
Regulatory		
Safety Standards	All parts must be fixed Structurally sound design	Design Team Testing Session
End of life Disposal	Plastic can be recycled Metal can be reused	Design Team
Codes of Practice	Conformed to BS 8888	Design Tutor

2.2 DESIGN CONCEPTS

The three main concepts considered are depicted in Table 3, alongside their main characteristics.

Table 3: Concepts and Considerations

DRIVE SYSTEM	SCHEMATIC REPRESENTATION	CONSIDERATIONS
PISTON	 *Figure 1: Piston concept*	• High precision is needed in the manufacturing process • High torque • A crankshaft is needed to translate linear to rotational motion • Material: aluminium or steel
TURBINE	 *Figure 2: Turbine concept*	• 3D printable • Material: ABS • Maximum diameter fixed if a direct drive approach is chosen

DIRECT PROPULSION	 *Figure 3: Direct drive nozzle concept*	• Less manufacturing required • Thrust provided might not be sufficient • Material: aluminium or steel

2.3 PROPULSION CHOICE

Pistons require very low tolerances that are difficult to achieve with available tools in the STW and within the time allocated for manufacture. There are also many moving parts required, which may require more time than is available for manufacture and can also lead to lower efficiencies compared to simpler designs. Pistons also need to be manufactured out of metal to achieve the low tolerance required, contributing a significant amount of weight. The disadvantages of this driving concept suggest that a different approach will be better.

The nozzle in direct propulsion is relatively simple to design or purchase. However, it is expected not to be able to complete the course.

Turbines can be easily 3D printed, especially for simpler geometries as simple turbines are not sensitive to small differences from high tolerance. Printing the component allows for a freedom of shapes and component designs that makes turbines the best candidate for our purpose. Printing a turbine in ABS also allows for very low weight compared to other solutions.

2.4 PROPULSION DESIGN DEVELOPMENT

Two main types of turbine design were considered - impulse and reaction, as depicted in Figures 3 and 4. Impulse turbines were chosen in the end due to their simpler design compared to the other two options, allowing for ease of manufacture and lower overall weight. The simplicity also reduced the overall weight of the turbine assembly.

Within impulse turbines, three different types were considered – Pelton, Turgo and crossflow. A simplified Pelton design was selected as it is less sensitive to nozzle positioning than Turgo and crossflow turbines. As most Pelton wheel designs are optimised for water, it was simplified and adapted to work with compressed gas instead, as detailed in Section 3.1**.**

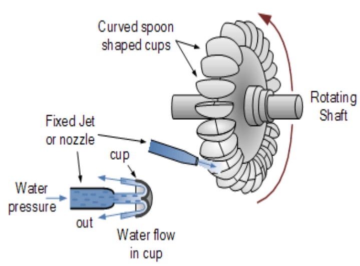


Figure 4 - Pelton impulse turbine (Adenwala, 2018)

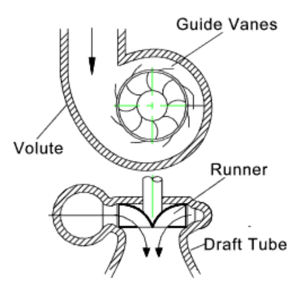


Figure 5 - Reaction turbine (Civil Engineering Terms, 2012)

2.5 POWER TRANSMISSION

Belt drive, gear drive and direct drive were considered for power transmission. Belt and gear drive allows for an increase in torque proportional to the gear ratio but add weight to the vehicle overall. The belt drive also allows for more flexibility in transmission positioning. However, the turbine design produces enough torque to propel the car at a satisfactory speed when attached directly to the real shaft. This is shown in sections 3.1 and 4.

Therefore, a direct drive transmission was chosen as it is much lighter and simpler to manufacture than any other transmission method.

2.6 FINAL DESIGN SUMMARY

The final design choices were based on two main criteria: manufacturing simplicity and weight. The chosen transmission system is direct drive with a turbine mounted directly on the rear shaft. The turbine receives air from a plastic hose connected to the pressure vessel and a nozzle is integrated in the turbine cover. The vehicle is within the ranges for dimensions and weight stated by the PDS.

3. FUNCTIONAL AND OPERATIONAL DESCRIPTION

3.1 TURBINE DESIGN

As air is much less viscous than water (0.01803 mPa/s vs 1.0518 mPa/s (The Engineering Toolbox, 2003)), it behaves very differently. Since Pelton wheel designs are optimised for water, it needed to be modified for use with air. The 3D printing machines available also have low resolution and high tolerance, thus more complicated surface designs may not correlate to theoretical calculations. A simple scoop shape was therefore used as the final design, as shown in Figure 6. To maximise the area that the compressed air will be incident to, a rectangular shape was used for the scoop. As a direct drive drivetrain was selected, the diameter of the turbine was restricted to the diameter of the rear wheels, 80 mm. To leave adequate clearance for the turbine housing, the turbine has an end-to-end diameter of 65 mm. This design is able to produce the required torque for the car to move off as shown in Section 4.

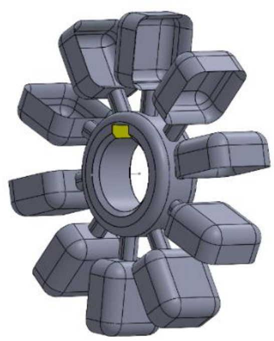


Figure 6 - Turbine CAD model

3.2 TURBINE COVER AND NOZZLE DESIGN

The turbine cover was designed to fit the turbine and minimize air leakage. The nozzle is built into the turbine cover inlet. The hose is cut to fit the distance between the valve and inlet and tightened by a hose clip to prevent air leakage (as shown in Figures 7 and 8).

The built-in converging nozzle has an outlet-inlet area ratio of 1/16, as shown in Figure 8. The nozzle converges the airflow and increases the momentum of air.

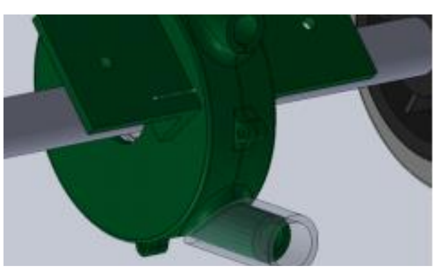


Figure 7 - Hose inlet connection

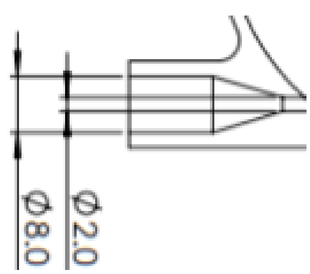


Figure 8 - Nozzle design and dimensions

Assuming the air is a perfect gas, the flow is inviscid, compressible, isentropic, and steady, maximum mass flow rate is given by Equation 1.

$$\dot{m_{max}} = P_0 A\sqrt{\frac{\gamma}{RT}}\left[\frac{2}{\gamma+1}\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{1}$$

The outlet velocity is given by equation 2,

$$\frac{A}{A_O} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{2}$$

Mach number and speed of sound is given by equation 3, assuming γ = 1.4.

$$M = \frac{v}{\sqrt{\gamma RT}} \tag{3}$$

The momentum is thus given by equation 4.

$$p_{max} = \dot{m_{max}} \cdot v \tag{4}$$

The nozzle increases the momentum of airflow per second to 1.32kg·m/s.

3.3 REAR SHAFT DESIGN

As the rear wheels are the outermost component on the shaft, the minimum rear shaft diameter was fixed by the diameter of the rear wheel bore, 12.3 mm.

Features of the shaft were designed to ensure functionality and ease of manufacture/assembly. For example, steps in the shaft diameter act to locate crucial components, and keys are used to ensure that parts rotate with the shaft. Fillets on any corners of shafts reduce stress concentrations and are a simple way to reduce the risk of failure.

Mild steel was chosen due to its ease of machining and higher tensile strength compared to aluminium.

3.4 BEARING AND BEARING HOUSING SELECTION

As the bearings would have minimal axial load, deep groove ball bearings were selected. Through calculations using torque estimations from the turbine, the pillow block housing units available were rated for much higher loads than the design required. Hence the extra weight of a large bearing unit was not necessary, and a simple housing made using CNC would be sufficient.

The wheels had an internal diameter of 12.3 mm, hence the next standard bearing size above 12.3 mm was chosen (15mm internal diameter, SKF 61902).

3.5 REAR SHAFT SUBASSEMBLY

Careful consideration was made to ensure the rear shaft subassembly was fully constrained. The bearing arrangement used is represented in Figure 9, where the end cap at the right is a representation of the spacer between the bearing and wheel in Figure 10. Acrylic washers are bolted to the outside of the bearing housing to act as internal circlips. The inner surface on the left is left floating, and with all surfaces on the right being constrained, the model is fully constrained overall. A keyway is used to constrain the turbine radially, and two external circlips are used to constrain it axially.

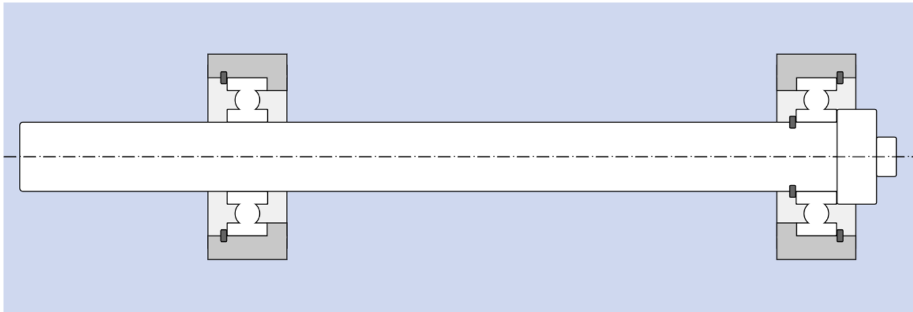


Figure 9 - Bearing Constraints Diagram

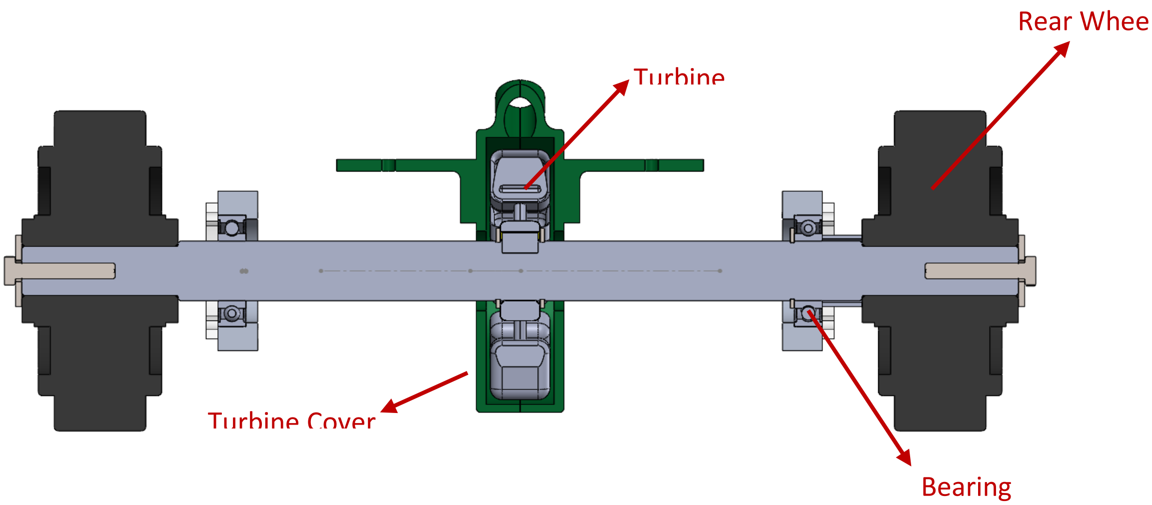


Figure 10 - Rear shaft subassembly section

4. EVIDENCE OF PERFORMANCE SATISFACTION

4.1 BEARING LIFE

The PDS sets the performance requirements as a minimum of 100 hours of life and the ability to travel the concourse length (assumed to be 20 ± 2m). The bearings were assumed to be most likely to fail first, as they carry most of the chassis' weight. Due to the bearing selection being made following a dimension criterion rather than based on load, the service life will likely be significantly higher than required. To evaluate the bearing's life, Equation 5 was used, with the average velocity *n* as 90 RPM.

$$L_{10} = \frac{C}{F} * 10^6/(n * 60) \text{ [hrs]} \tag{5}$$

The basic dynamic load rating, *C,* for the selected bearing is 4.36 kN, and hence the expected life is estimated to be $3.7 * 10^{10}\ hrs$.

4.2 TURBINE TORQUE

To estimate the distance travelled when the canister valve is opened, a few steps were taken. Firstly, the initial torque provided by the turbine to the rear shaft was calculated. It was compared to the static friction between the wheels and the ground to check if the torque provided is sufficient for the vehicle to start moving. The torque required to start the motion was evaluated to be 0.33 Nm. The torque provided by the turbine was found to be 0.42 Nm using the result from Equation 6, then torque on the wheels was evaluated with Equation 7.

$$F_{wheel} = \frac{q^2}{\rho * A_2} \tag{6}$$

$$T_{wheel} = \frac{D_{wheel}}{D_{tur}} T_{tur} \quad (7)$$

Where q is the mass flow rate, and A_2 is the cross-sectional area of the air stream at the turbine inlet. As the torque provided is more than the required torque, the vehicle will be able to move off with a direct drive system.

4.3 DISTANCE TRAVELLED

A force balance was then performed on the wheels to evaluate the value of angular acceleration for a set of small-time intervals, shown in the equation below (Eq. 8).

$$\alpha = \frac{(T_{wheel} - F_f * r)}{I} \quad (8)$$

From the angular acceleration, the angular velocity can be calculated with the Equation 9, then converted into the car's linear velocity.

$$\omega_{n+1} = (\alpha_n * \delta t_n) + \omega_n \quad (9)$$

The set of linear velocities is associated with a set of time intervals and multiplying the two values gives the displacement during each time interval. The total displacement can be found by adding the instantaneous displacement values in Equation 10.

$$\sum_{n=0}^{k} (v_n * \delta t_n) \quad (10)$$

Displacement against time is shown in Figure 12 and it results in an expected distance travelled of 18 meters, matching the length estimate of the concourse.

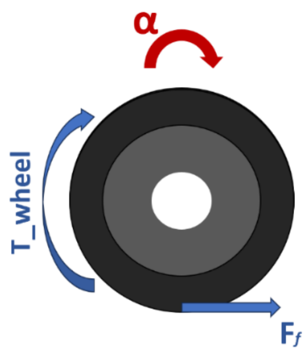


Figure 11: Free body diagram of a rear wheel

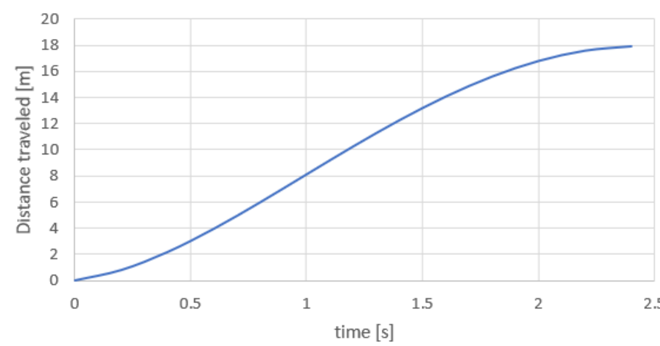


Figure 12: Vehicle Displacement

5. FAILURE MODES

5.1 STRESS ANALYSIS

The rear shaft stress analysis was carried out to investigate the possible failure of the component when operating under normal conditions.

The maximum stresses on the shaft are expected to be experienced at the beginning of the operating period when the valve has just been opened. The minimum diameter of the shaft is 12.3 mm, and it is assumed to be uniform for the whole shaft in order to simplify calculations. As there is only one step to 15 mm, the assumption is valid. The result will also overestimate the maximum stresses and deflections, resulting in a higher actual safety factor.

The three-dimensional free-body diagram of the shaft is shown in Figure 14, where R_w is the reaction force from the wheel, W_{ch} is half of the weight of the chassis, W_{tur} is the weight of the turbine, W_{sh} is the weight of the shaft, T_{tur} is the force that the turbine exerts on the shaft, and F_b is the reaction force to balance the latter from each bearing. All the forces are assumed to be point forces and the two main planes are taken into consideration (horizontal and vertical). The magnitudes of the forces are shown in Table 4.

Table 4: Loads present on the rear shaft

W_{tur}	W_{ch}	W_{sh}	R_w	T_{tur}	F_b
-0.69 N	-6.37 N	-1.03 N	7.24	-8 N	4 N

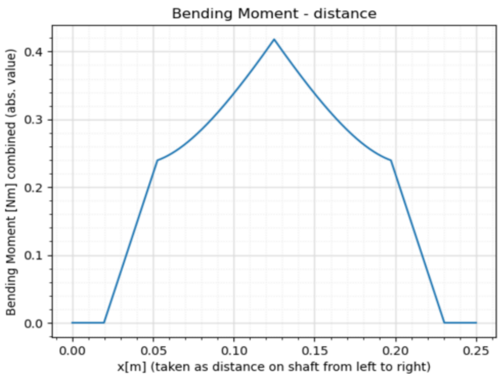


Figure 13 - Combined Bending Moment

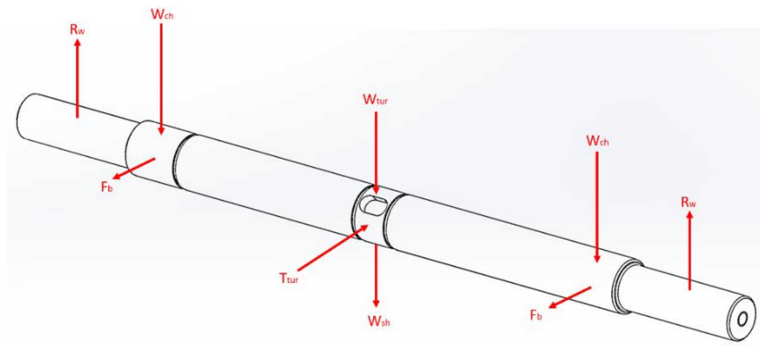


Figure 14: Free Body Diagram of Rear Shaft

The bending moments in the two planes were combined using equation 11 to find the combined bending moment shown in Figure 13.

$$M_{combined} = \sqrt{|M_{horiz}|^2 + |M_{vertic}|^2} \tag{11}$$

The values of bending moments in the two planes were then used to calculate the direct stresses in each direction using Equation 12.

$$\frac{\sigma}{y} = \frac{M}{I} \tag{12}$$

Using the maximum value of torque provided to the shaft, the shear stress τ was calculated from Equation 13.

$$\frac{\tau}{r} = \frac{T}{J} \tag{13}$$

The direct stresses and the shear stress were used to find the maximum values of direct stresses from Equation 14 and confirmed by Mohr's Circle as shown in Figure 15. The values of σ_1 and σ_2 are 3.17 and 0.05 MPa.

$$\sigma_{1,2} = \frac{1}{2}[(\sigma_x + \sigma_y) \pm \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2} \tag{14}$$

The results were compared to the maximum allowable stress for mild steel, to find a minimum safety factor in the shaft of 75. The SF is significantly higher than needed, meaning that the shaft design will not fail under stress. This high SF could be reduced by reducing the shaft diameter. However, the minimum value for this feature is set by the wheels' internal diameter (12.3 mm) and by the higher following standard shaft diameter (15 mm).

The acrylic chassis will have significant deflection towards the middle due to the low elastic modulus of elastic and length of the chassis. However, it does not result in deformation and does not affect the ground clearance of the turbine housing and can thus be disregarded.

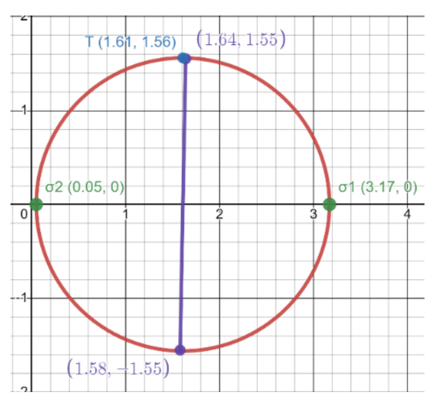


Figure 15: Mohr's Circle

6. MANUFACTURING

The manufacturing of the parts was carried out over five sessions in the Student Teaching Workshop. Three processes were performed for us, namely additive manufacturing, CNC, and acrylic laser cutting, while each team member worked on the manufacturing involving drilling, milling and turning. The details of manufacturing for each of the components are summarised in Table 5 below. A detailed measurement report on the rear shaft and bearing housing is given in Appendix A.

An advantage of a simple design meant fewer components to be manufactured and a simpler assembly process. Because of this, the group was not under significant time pressure and could spend more time on high quality manufacturing. This minimises the potential points of failure for the vehicle. Overall, the group was able to successfully manufacture and assemble the vehicle to a high quality. Additional time scheduled in the STW was used to test performance, identify limiting factors, and make minor changes to components. For example, additional time was used to improve the surface finish of the turbine to reduce friction and improve performance.

Table 5 - Manufacturing Details of Main Parts

COMPONENT	PROCESS	MACHINE	MATERIAL	CONSIDERATIONS
General	-	-	-	Fillets and chamfers were added to most sharp edges
Spacer	Turning / Drilling	Lathe / Drill	Mild Steel EN1A	Not subject to significant stresses
Front blocks	Drilling - Milling	Drill – Milling Machine	Aluminium 6082 T6	Min radius of 3 mm
Bearing Housing	CNC	CNC	Aluminium 6082 T6	Min radius of 3 mm
Shaft	Turning	Lathe	Mild Steel EN1A	Most complex part to manufacture due to the number of features and to the precision needed
Turbine and case	Additive manufacturing	3D printer	ABS	Accuracy and flexibility of customisation
Chassis	Laser cutting	Laser cutter	Acrylic	Light weight

7. FINAL ASSEMBLY

Figure 16 represents the six stages of assembling. The first step is to insert the turbine in the turbine case, fastening the two parts of the cover together. The turbine-cover subassembly is then fastened in its final position to the acrylic base. One bearing and its relative housing are fastened to the acrylic base and the shaft with the key is slid in, firstly passing in the turbine bore (paying attention to the turbine that is slightly misaligned due to clearance inside the cover) and then through the bearing until the shaft’s shoulder touches the bearing's inner ring. The turbine is constrained using circlips on both sides. The wheel on the bearing side can be then mounted and fastened, while the other bearing and housing are fastened and constrained properly.

Then, the spacer and the other wheel are inserted into the assembly. The air canister, the valve, the connector, and the hose are connected to the turbine inlet and fastened to the acrylic base. Lastly, the two mounting blocks and the front wheels are fastened to the final assembly.

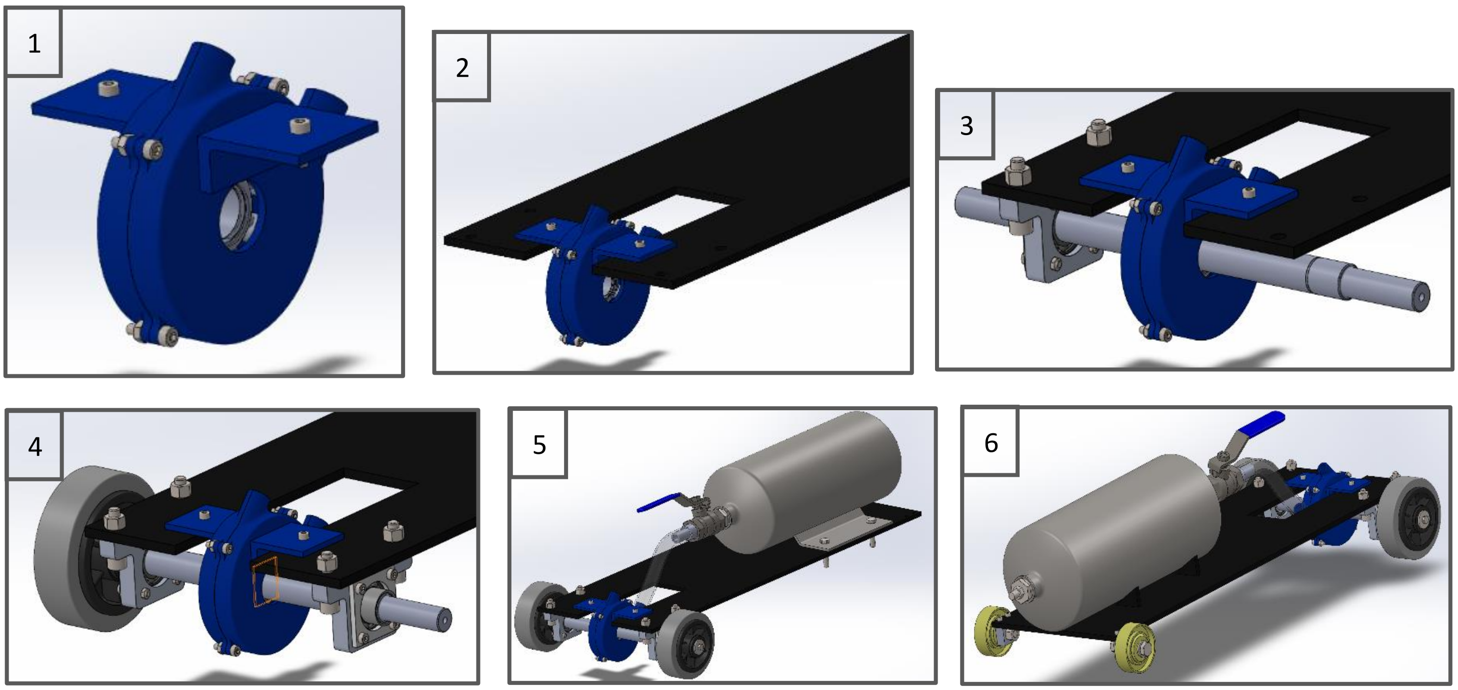


Figure 16: Assembly order of vehicle, going left to right

8. CONCLUSIONS

Our group was tasked with the design and manufacture of a compressed air-powered car, which would be raced against other groups on a 20 m track. The design had to meet some critical satisfaction criteria; these included 100 hours of operational life, using a given air canister and wheels, and external parts being limited to one supplier (RS components). Aside from these constraints, the group was given complete design freedom as to the design and operation of the vehicle.

In the early stages of design various concepts were considered. These were direct propulsion, piston-driven, and turbine propulsion. After careful consideration including some rough calculations, the group decided to create a design revolving around the use of a turbine to drive the wheels. As a group, we decided on some important characteristics of the design, such as being lightweight and simple. With these considerations we were able to design and manufacture a direct drive turbine vehicle, achieving simplicity and lightweight by removing any transmission components (gears, pulleys, extra shafts) which we deemed unnecessary.

All decisions made from early concept to final design were informed by various calculations. These include estimations of torque required, mass flow calculations from a nozzle, and stress analysis of components such as the rear shaft.

The simple design aided manufacturing ability greatly. With a minimal part list and simple assembly process, more time could be spent on the manufacturing of each part, ensuring high-quality part production. The limited time pressure also meant that small changes could be discussed and implemented during the manufacturing sessions. In the end, we were able to manufacture and assemble our vehicle with high accuracy to the original design, and with the functionality we predicted.

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APPENDIX A – MEASUREMENT REPORT

The following tables 6 and 7 present the data gathered measuring two critical components: the rear shaft and the bearing housing. The two reference drawings are P40_2P002 and P40_RS002 in Appendix C and the Dimension IDs follow a numerical order and are labelled L or D based on the nature of the measure: L for lengths and D for diameters.

Table 6 - Rear shaft bearing housing measurement report

Part Number: P40_2P002				
Title/Description: Rear shaft bearing housing				
Dimension ID	Value and Tolerance/mm	Measurement Tool	Measurement/mm	Conclusion
H1	40±0.5	Vernier Caliper	40.01±0.005	Conform to tolerance
H2	10±0.5	Vernier Caliper	10.10±0.005	Conform to tolerance
T1	10±0.5	Vernier Caliper	9.99±0.005	Conform to tolerance
T2	7±0.5	Steel Ruler	7.0±0.05	Conform to tolerance
T3	10±0.5	Vernier Caliper	10.03±0.005	Conform to tolerance
D1	Ø26±0.5	Vernier Caliper	26.05±0.005	Conform to tolerance
D2	Ø28±0.5	Vernier Caliper	27.98±0.005	Conform to tolerance
HD1	Ø3.4±0.05	Vernier Caliper	3.37±0.005	Conform to tolerance
HD2	Ø6.6±0.05	Vernier Caliper	5.07±0.005	Does not fit tolerance, but bolt fit the hole and constraint the housing.
L0	70±0.5	Vernier Caliper	69.96±0.005	Conform to tolerance
L1	40±0.5	Vernier Caliper	39.95±0.005	Conform to tolerance
L11	13±0.5	Vernier Caliper	12.97±0.005	Conform to tolerance
L12	13±0.5	Vernier Caliper	13.00±0.005	Conform to tolerance
L13	26±0.5	Vernier Caliper	26.12±0.005	Conform to tolerance
L14	26±0.6	Vernier Caliper	25.93±0.005	Conform to tolerance
L15	26±0.7	Vernier Caliper	26.02±0.005	Conform to tolerance
L16	26±0.8	Vernier Caliper	26.00±0.005	Conform to tolerance
L21	20±0.5	Vernier Caliper	19.88±0.005	Conform to tolerance
L22	20±0.6	Vernier Caliper	19.99±0.005	Conform to tolerance
L31	5±0.5	Vernier Caliper	5.09±0.005	Conform to tolerance
L32	7.5±0.5	Vernier Caliper	7.46±0.005	Conform to tolerance
L33	55±0.5	Vernier Caliper	54.79±0.005	Conform to tolerance
L34	5±0.5	Vernier Caliper	5.05±0.005	Conform to tolerance

Table 7 - Rear Shaft Measurement Report

Part Number: P40_RS002				
Title/Description: Rear shaft				
Dimension ID	Value and Tolerance/mm	Measurement Tool	Measurement/mm	Conclusion
L0	250.0±0.5	Metre ruler	250±0.5	Conforms to tolerance
L1	39.20+0.15	Vernier Caliper	39.30±0.005	Conforms to tolerance
L2	118.90+0.20	Vernier Caliper	188.80±0.005	Conforms to tolerance
L3	39.20+0.15	Vernier Caliper	39.22±0.005	Conforms to tolerance
L4	56.20+0.20	Vernier Caliper	56.20±0.005	Conforms to tolerance
L5	118.90+0.20	Vernier Caliper	119.00±0.005	Conforms to tolerance
L6	8.0±0.5	Vernier Caliper	8.00±0.005	Conforms to tolerance
L7	22.5±0.5	Vernier Caliper	22.47±0.005	Conforms to tolerance
L8	22.5±0.5	Vernier Caliper	22.46±0.005	Conforms to tolerance
R1	Ø3.3±0.5	Vernier Caliper	Ø3.30±0.005	Conforms to tolerance
R2	Ø12.3±0.5	Vernier Caliper	Ø12.26±0.005	Conforms to tolerance
R3	Ø15.0±0.5	Vernier Caliper	Ø14.80±0.005	Conforms to tolerance
R4	Ø3.3±0.5	Vernier Caliper	Ø3.50±0.005	Conforms to tolerance
R5	Ø12.3±0.5	Vernier Caliper	Ø12.35±0.005	Conforms to tolerance
D1	3.0±0.5	Metal ruler	3.1±0.5	Conforms to tolerance
D2	4.0±0.5	Vernier Caliper	3.96±0.005	Conforms to tolerance
D3	5.00-0.04	Vernier Caliper	5.01±0.005	Does not fit tolerance, but key fit in
D4	3.0±0.5	Vernier Caliper	2.96±0.005	Conforms to tolerance
D5	1.0±0.5	Vernier Caliper	1.01±0.5	Conforms to tolerance
H1	12.00-0.03	Vernier Caliper	11.95±0.005	Conforms to tolerance
H2	0.43-0.11	Vernier Caliper	0.40±0.005	Conforms to tolerance

APPENDIX B – INDIVIDUAL TIME LOG AND GANTT CHART

Table 8 is a summary of the hours spent individually on the project with a rough estimate of the breakdown according to each task. Figure 17 is a Gantt Chart to show the allocation of time throughout the length of the project.

Table 8- Individual time log

MEMBER	DESIGN AND MEETINGS	PERFORMANCE EVALUATION	CAD	REPORT	MANUFACTURING	TOTAL
Matthew Morrissey	15	10	20	22	6	73
Micheal Pan	25	10	25	30	6	96
Gian Andrea Rossi	19	18	17	42	6	102
Grady Su	15	10	17	14	6	62
Tanadon Tan Tharahirunchot	19	9	6	14	12	60

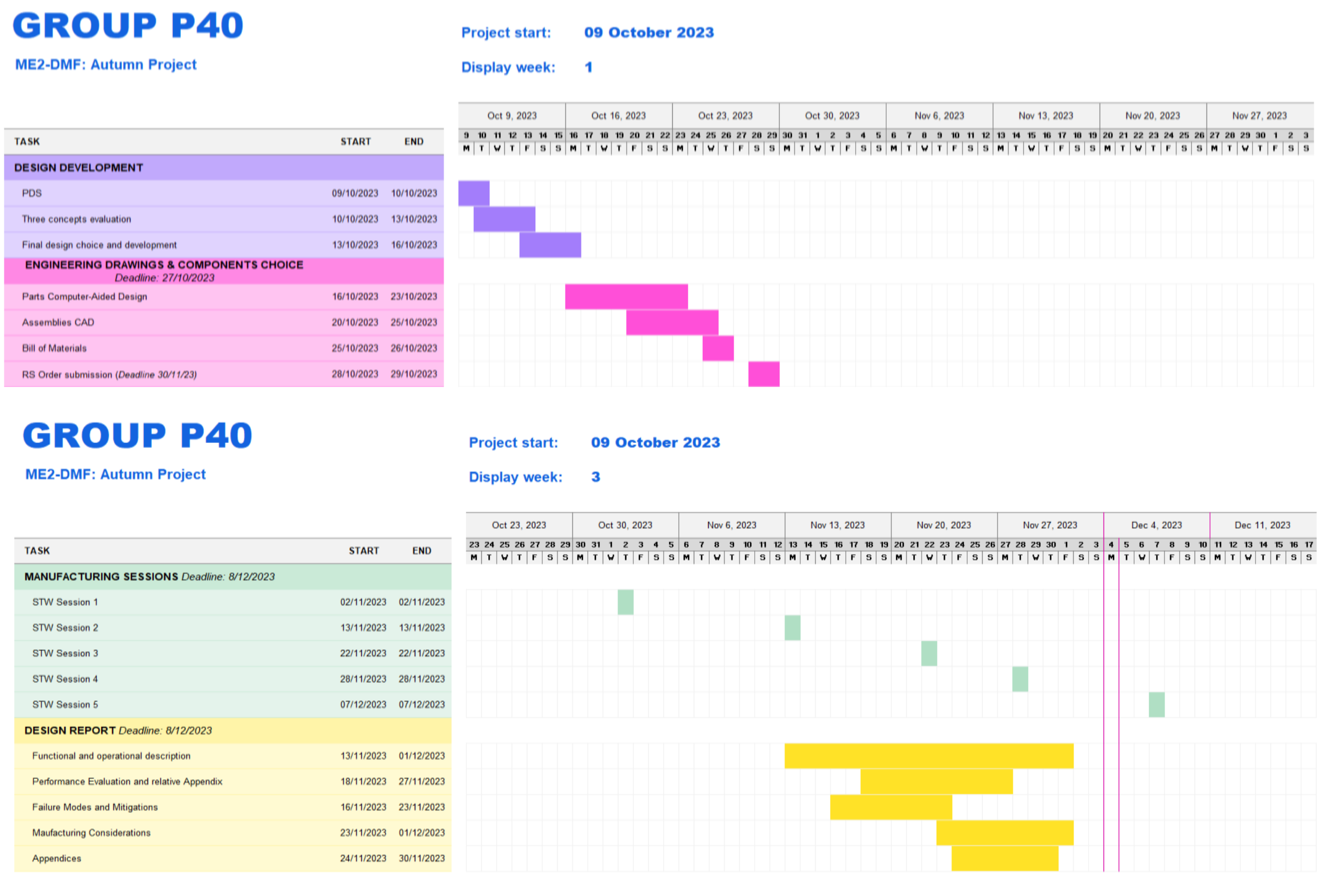
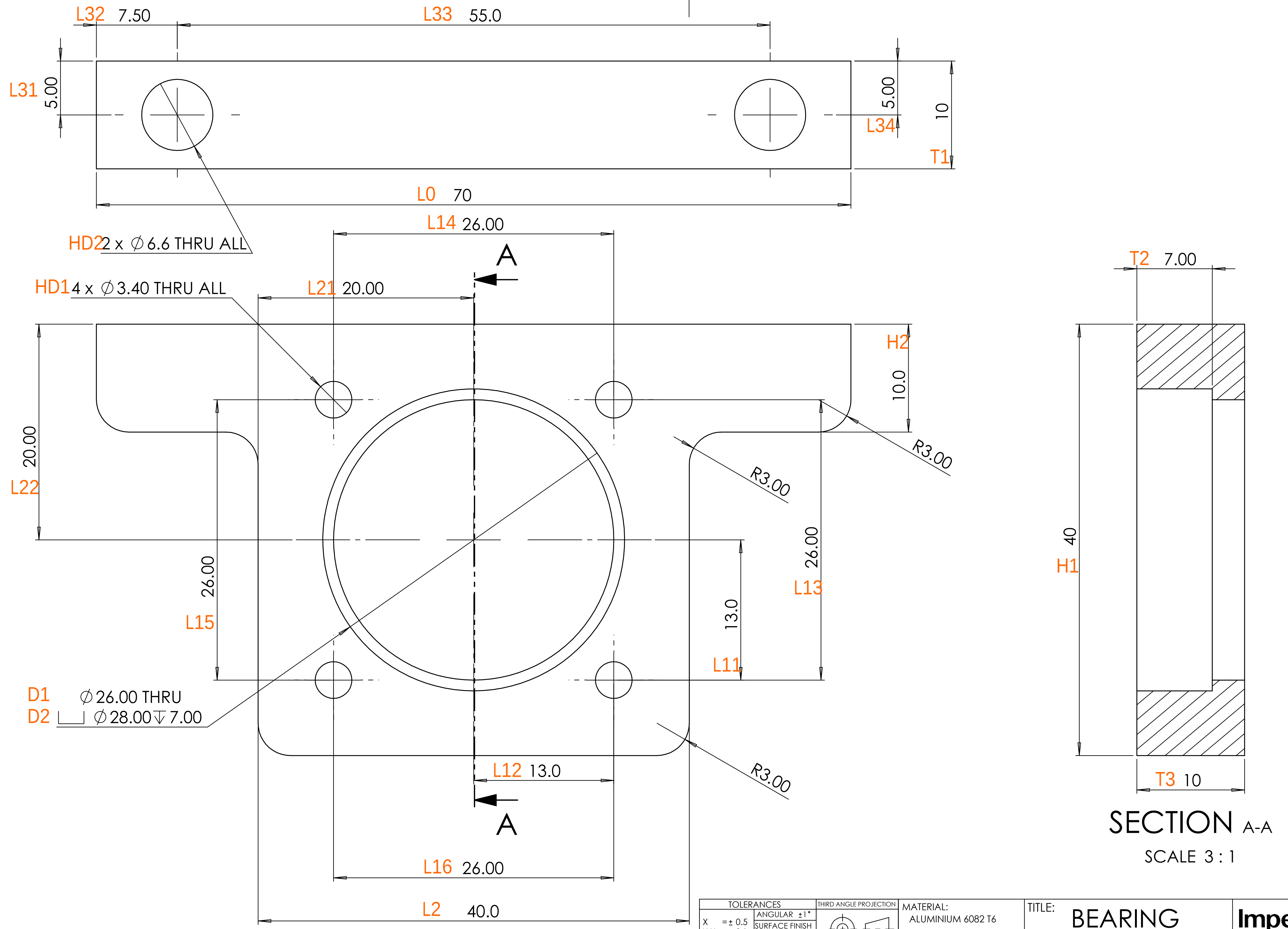


Figure 17: Gantt Chart

APPENDIX C – ENGINEERING DRAWINGS

The next pages show the engineering drawings for the Bearing housing, the rear shaft, and the rear shaft assembly in this order. Followed by the General Assembly drawing and its relative bill of Materials.

Discussing with Dr. Ambrose Taylor, the engineering drawings choice disagrees with the guidelines: the design includes simple parts, therefore seems irrelevant to include a part drawing that shows little engineering skills instead of a subassembly drawing. After this consideration we agreed to include two parts, namely the bearing housing ad the shaft, and to substitute the third part drawing with the rearshaft subassembly drawing, that shows important details for the assemblage of the final product, aswell as the Bill of Materials for the subassembly.



TOLERANCES: X = ± 0.5, X.X = ± 0.1, X.XX = ± 0.02; ANGULAR ± 1°; SURFACE FINISH MACHINED FACES Ra 6.3	THIRD ANGLE PROJECTION	MATERIAL: ALUMINIUM 6082 T6	TITLE: BEARING HOUSING	Imperial College London Department of Mechanical Engineering
		ALL DIMENSIONS ARE IN MILLIMETRES		
	NAME	DATE		
DRAWN	MICHAEL PAN	10/11/23	DO NOT SCALE DRAWING	DWG No. P40_2P002
CHECKED	GIAN ANDREA ROSSI	10/11/23		
APPROVED	MATTHEW MORRISSEY	10/11/23	A3 SCALE 3:1	SHEET 1 OF 1 REVISION 2

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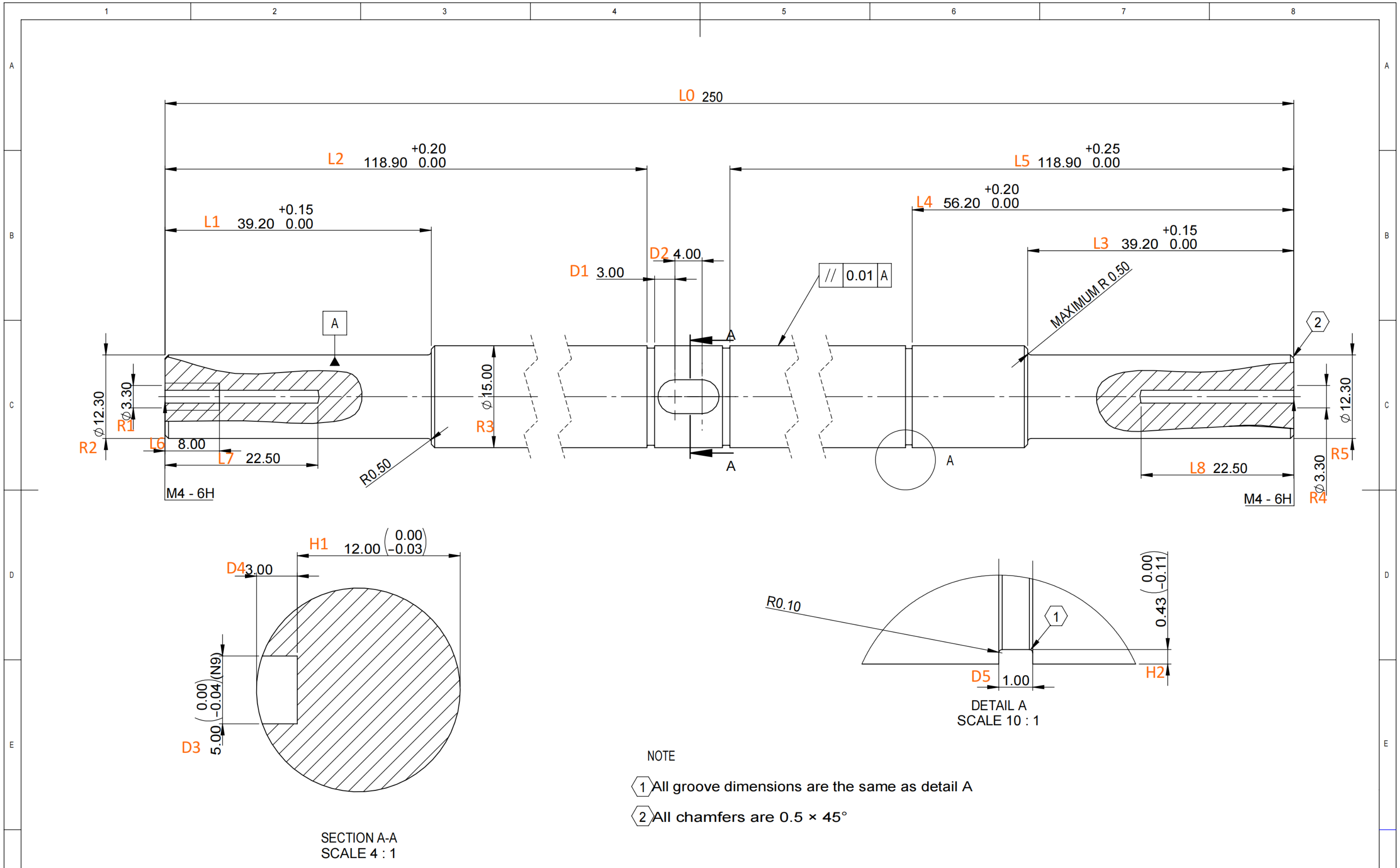
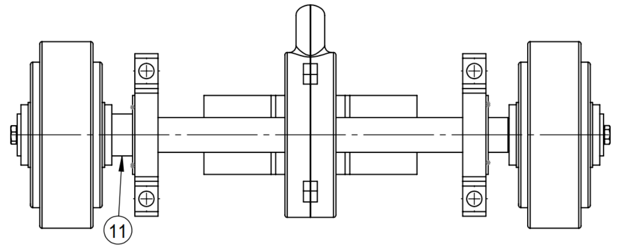
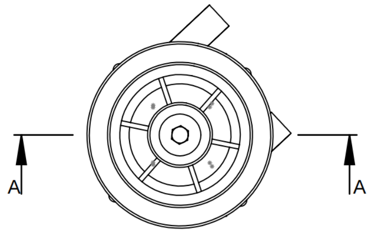
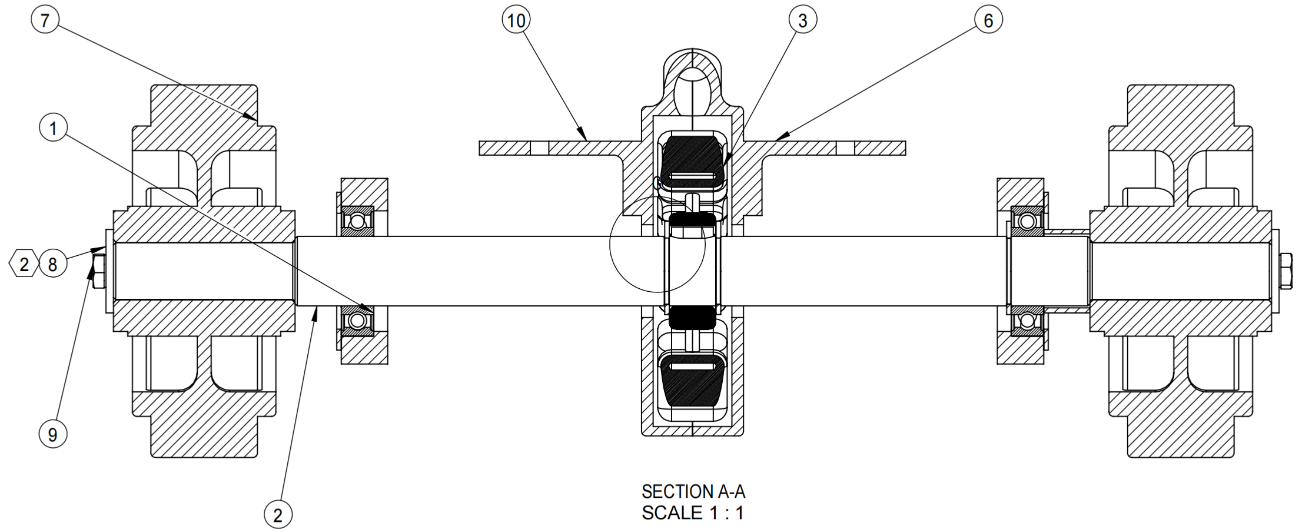


TOLERANCES X = ± 0.5 X.X = ± 0.1 X.XX = ± 0.02	ANGULAR ±1° SURFACE FINISH MACHINED FACES Ra 6.3	THIRD ANGLE PROJECTION	MATERIAL: MILD STEEL ALL DIMENSIONS ARE IN MILLIMETRES	TITLE: Rear Shaft	Imperial College London Department of Mechanical Engineering
	NAME	DATE	DO NOT SCALE DRAWING	DWG No.	
DRAWN	GRADY SU	30/10/2023	A3	P40_RS002	SHEET 1 OF 1
CHECKED	MACHEAL PAN	30/10/2023	SCALE 1:2		REVISION
APPROVED	MATT MORRISSEY	1/11/2023			

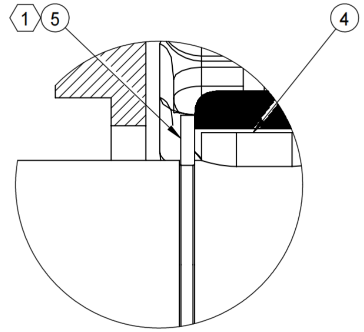
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	P40_BR001	REARWHEEL BEARING	2
2	P40_RS002	REARWHEEL SHAFT	1
3	P40_TF001	TURBINE FAN	1
4	P40_KEY	TURBINE KEY	1
5	P40_BS_3673-4	TURBINE CIRCLIPS	3
6	P40_TC001	LEFT TURBINE COVER	1
7	P40_RW001	REARWHEEL	2
8	P40_SW-10669	REARWHEEL WASHER	2
9	P40_ST_M4	M4 X 25 STUD	2
10	P40_TC002	RIGHT TURBINE COVER	1
11	P40_SP001	REARWHEEL SPACER	1







SECTION A-A
SCALE 1 : 1



DETAIL G
SCALE 3 : 1

1. Circlips and Key used to add axile and vertical constraint to the turbine
2. A washer and 4 studs into the wheel hub add vertical and axile constraint to the rearwheel

TOLERANCES
X = ± 0.5
X.X = ± 0.1
X.XX = ± 0.02
ANGULAR ±1°
SURFACE FINISH MACHINED FACES Ra 6.3

THIRD ANGLE PROJECTION

	NAME	DATE
DRAWN	GRADY SU	31/10/2023
CHECKED	MATT MORRISSEY	1/11/2023
APPROVED	TANADON T	1/11/2023

MATERIAL:

ALL DIMENSIONS ARE IN MILLIMETRES

DO NOT SCALE DRAWING

A3 SCALE 1:2

TITLE: REAR SHAFT ASSEMBLY

DWG No. P40_A002

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Department of Mechanical Engineering

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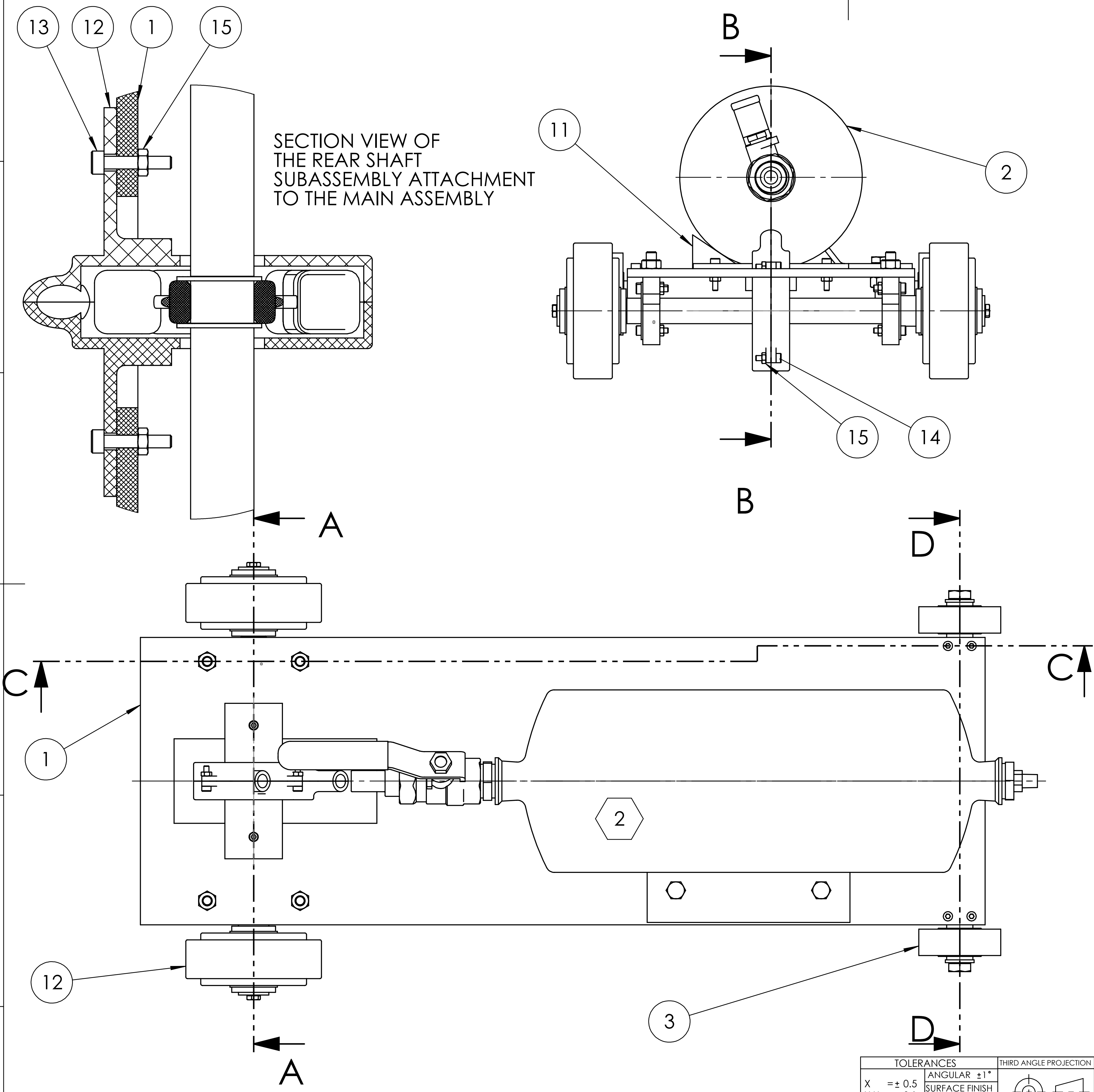

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	P40_1P003	ACRYLIC BASE	1
2	P40_A002	AIR CANISTER SUBASSEMBLY	1
3	P40_1P007	FRONT WHEEL	2
4	P40_1P004	FRONT WHEEL SUPPORT BLOCK	2
5	P40_WSH_M8	WASHER BS4320 M8	2
6	P40_XBLT_M8X40	HEX BOLT M8 X 40	2
7	P40_XBLT_M6X30	HEX BOLT M6 X 30	2
8	P40_XNUT_M8	NUT M8	2
9	P40_XNUT_M6H5	NUT M6	2
10	P40_XBLT_M3X10	HEX BOLT M3 X 10	4
11	P40_1P006	CANISTER SUPPORT	2
12	P40_A002	REAR SHAFT SUBASSEMBLY	1
13	P40_XBLT_M3X10	HEX BOLT M3 X 16	2
14	P40_XBLT_M3X12	HEX BOLT M3 X 12	4
15	P40_XNUT_M3	HEX NUT M3	6
16	P40_XBLT_M3X10	HEX BOLT M6 X 25	4
17	P40_2P008	RS CONNECTOR - SIMPLIFIED REPRESENTATION	1
18	P40_XNUT_M8H6	HEX NUT M6	4

TOLERANCES
X = ± 0.5
X.X = ± 0.1
X.XX = ± 0.02
ANGULAR ±1°
SURFACE FINISH MACHINED FACES Ra 6.3
THIRD ANGLE PROJECTION

MATERIAL: N/A
ALL DIMENSIONS ARE IN MILLIMETRES
DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	GIAN ANDREA ROSSI	26/10/23
CHECKED	MATTHEW MORRISSEY	26/10/23
APPROVED	MICHAEL PAN	31/10/23

TITLE: GENERAL ASSEMBLY

DWG No. P40_A000

A3 SCALE 2:5

Imperial College London
Department of Mechanical Engineering

SHEET 1 OF 3 REVISION 3

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